SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)
CATEGORY - 1 UNIVERSITY BY UGC
Accredited with 'A++' grade by NAAC | Approved by AICTE
www.sathyabama.ac.in

SCHOOL OF SCIENCE AND TECHNOLOGY

Department of Computer Application

Internship Project

DEEP LEARNING - BASED SKIN CANCER DETECTION USING DERMOSCOPIC IMAGE ANALYSIS

PRESENTED BY :

S . Jayaraj (44750020)

MS. Jai Dharun (44750018)

SUPERVISED BY :

Dr. J. Palanimeera

ABSTRACT

- Digital Skin cancer is one of the most common and life-threatening diseases, making early detection essential.
- This project uses deep learning and dermoscopic image analysis to detect skin cancer accurately.
- A Convolutional Neural Network (CNN) is used to automatically extract important features from skin lesion images.
- The trained model classifies skin lesions as Benign or Malignant with high accuracy.
- The proposed system supports dermatologists by providing faster, reliable, and efficient skin cancer diagnosis.

INTRODUCTION

- Skin cancer is one of the most common and dangerous types of cancer worldwide.
- Early detection is important to increase the chances of successful treatment.
- Dermoscopic images provide detailed information about skin lesions for accurate analysis.
- Deep Learning, especially Convolutional Neural Networks (CNN), can automatically detect and classify skin cancer.
- This project aims to develop an accurate and efficient skin cancer detection system using dermoscopic image analysis.

PROBLEM STATEMENT

- Manual skin cancer diagnosis requires experienced dermatologists and specialized equipment.
- Early-stage skin cancer is difficult to identify through visual examination alone.
- Human diagnosis may lead to errors due to variations in skin lesion appearance.
- Analyzing a large number of dermoscopic images manually is time-consuming.
- Delayed or inaccurate diagnosis can reduce the chances of successful treatment.
- Therefore, an automated deep learning-based system is required to provide accurate and early skin cancer detection.

OBJECTIVES

- To develop an automated system for detecting skin cancer using dermoscopic images.
- To accurately classify skin lesions as **benign** or **malignant** using deep learning techniques.
- To improve the accuracy of skin cancer diagnosis through Convolutional Neural Networks (CNN).
- To reduce the time and human errors involved in manual skin cancer detection.
- To assist dermatologists by providing reliable and efficient diagnostic support.
- To enable early detection of skin cancer, improving treatment outcomes and patient survival.

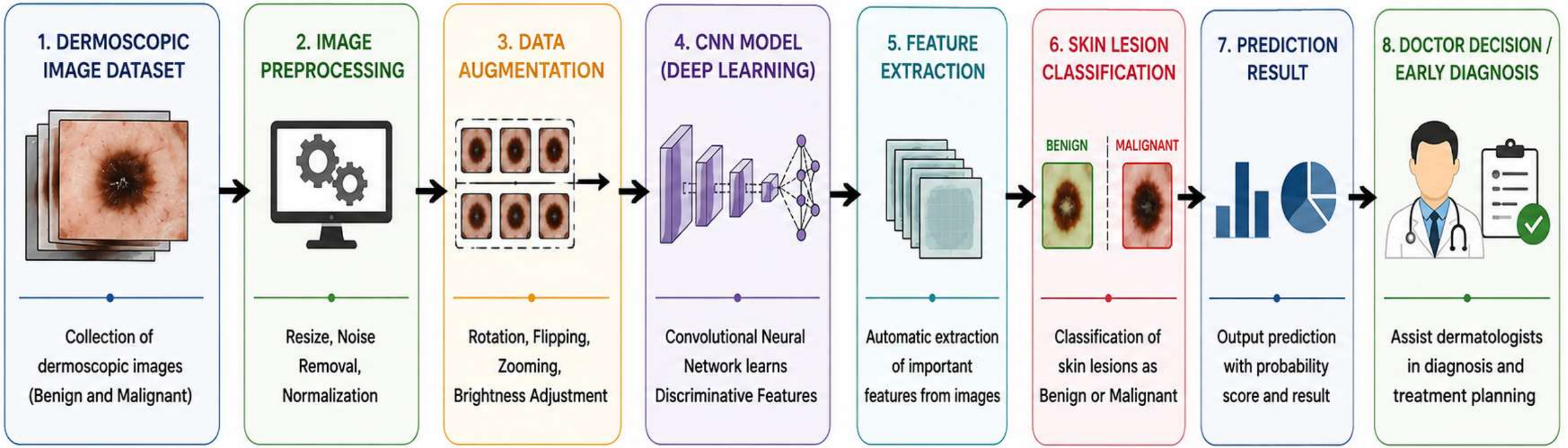
TECHNOLOGIES USED

- **Python** – Used for implementing the skin cancer detection system.
- **TensorFlow / Keras** – Used to build and train the CNN deep learning model.
- **OpenCV** – Used for image preprocessing and enhancement.
- **Convolutional Neural Network (CNN)** – Used to classify skin lesions as Benign or Malignant.
- **ISIC / HAM10000 Dataset** – Used for training and testing dermoscopic skin lesion images.
- **Flask, HTML, CSS & JavaScript** – Used to develop the web-based user interface for image upload and prediction.

EXISTING AND PROPOSED SYSTEM

EXISTING SYSTEM	PROPOSED SYSTEM
Uses CNN only for skin cancer classification.	Uses CNN integrated with Explainable AI (XAI) for transparent prediction.
Provides only the final prediction (Benign/Malignant).	Provides both prediction and explanation for each result.
Does not explain how the prediction is made.	Highlights important lesion features influencing the prediction.
Does not provide a confidence or risk score.	Generates a Risk Score (Low, Medium, High) based on lesion characteristics
Limited transparency reduces trust in automated diagnosis.	Improves transparency, reliability, and clinical decision support for dermatologists.

SYSTEM ARCHITECTURE



METHODOLOGY

- The dermoscopic skin image dataset is collected from reliable sources such as ISIC or HAM10000.
- The collected images are preprocessed by resizing, noise removal, normalization, and data augmentation.
- The preprocessed images are used to train a Convolutional Neural Network (CNN) model.
- The CNN model automatically extracts important features from the skin lesion images.
- The trained model classifies the skin lesions as Benign or Malignant.
- The final prediction result is displayed to assist dermatologists in accurate and early skin cancer diagnosis.

MODULE 1: IMAGE PREPROCESSING

Objective:

Prepare dermoscopic skin images by improving their quality before training the deep learning model.

Functions:

- Collect dermoscopic skin lesion images from datasets such as ISIC or HAM10000.
- Resize all images to a fixed size for uniform processing.
- Remove image noise and improve image quality.
- Normalize pixel values for better model performance.
- Apply data augmentation techniques such as rotation, flipping, and zooming.
- Generate high-quality images for CNN model training.

MODULE 2: SKIN CANCER CLASSIFICATION USING DEEP LEARNING

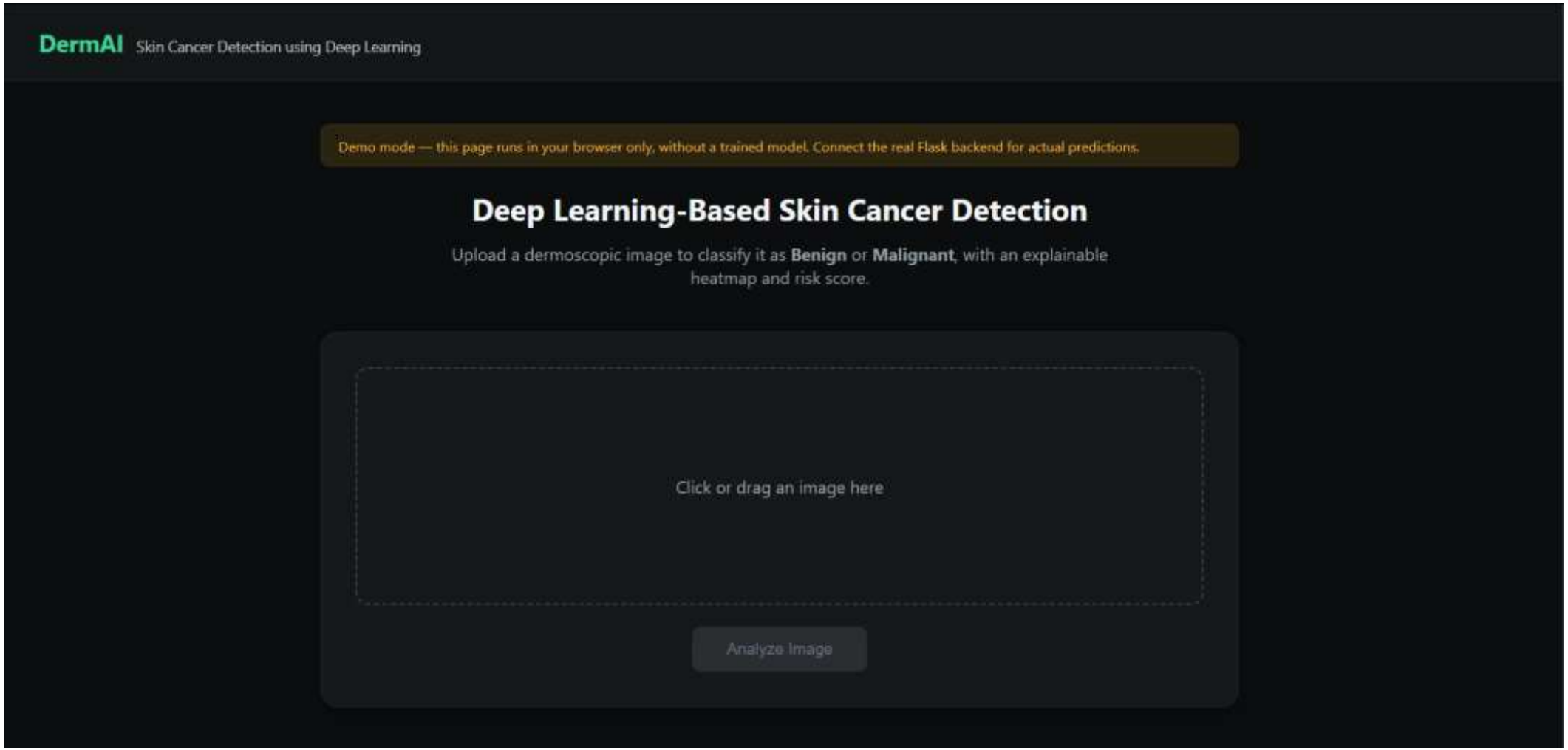
Objective:

Classify dermoscopic skin lesion images as **Benign** or **Malignant** using a Convolutional Neural Network (CNN).

Functions:

- Train the CNN model using preprocessed dermoscopic images.
- Automatically extract important features from skin lesion images.
- Learn the patterns of benign and malignant skin lesions.
- Classify skin lesions into benign or malignant categories.
- Predict the final diagnosis with high accuracy.
- Display the prediction result to support dermatologists in diagnosis.

IMPLEMENTATION - DEMO



ADVANTAGES

- Detects skin cancer at an early stage, improving the chances of successful treatment.
- Provides higher accuracy than traditional manual diagnosis methods.
- Reduces human errors by automatically analyzing dermoscopic images.
- Saves time by providing faster prediction and classification results.
- Assists dermatologists in making reliable clinical decisions.
- Can analyze large numbers of skin lesion images efficiently.



APPLICATIONS

- Used in hospitals for early detection and diagnosis of skin cancer.
- Assists dermatologists in analyzing dermoscopic images more accurately.
- Supports telemedicine services for remote skin disease diagnosis.
- Helps medical researchers study skin cancer using AI-based image analysis.
- Can be integrated into healthcare systems for automated skin lesion screening.
- Useful in medical colleges and research institutions for educational purposes.

FUTURE SCOPE

- Improve accuracy using advanced deep learning models.
- Develop a mobile application for real-time skin cancer detection..
- Integrate cloud technology for remote diagnosis.
- Expand the system to detect multiple skin diseases.
- Increase the dataset size for better model performance.



CONCLUSION

- The proposed system accurately detects skin cancer using deep learning and dermoscopic image analysis.
- The CNN model effectively classifies skin lesions as **Benign** or **Malignant**.
- The system reduces human errors and improves diagnostic accuracy.
- It supports dermatologists by providing faster and more reliable predictions.
- Early detection of skin cancer can improve treatment outcomes and increase patient survival rates.

REFERENCES

- *S. Shafiullah et al., "Detection of Skin Cancer: A Deep Learning Approach," in 2023 IEEE Smart World Congress (SWC), Portsmouth, UK, 2023, pp. 1–8, doi: 10.1109/SWC57546.2023.10448820.*
- *A. K. Arun and M. Palmer, "Skin Cancer Detection Using Deep Learning," in 2024 10th International Conference on Communication and Signal Processing (ICCSP), Chennai, India, 2024, pp. 1712–1717, doi: 10.1109/ICCSP60870.2024.10543954.*
- *A. Magdy, H. Hussein, R. F. Abdel-Kader, and K. A. E. Salam, "Performance Enhancement of Skin Cancer Classification Using Computer Vision," IEEE Access, vol. 11, pp. 72120–72133, 2023, doi: 10.1109/ACCESS.2023.3294974.*
- *H. Kahramanli Ornek, B. Yilmaz, E. Yasin, and M. Koklu, "Deep Learning-Based Classification of Skin Lesion Dermoscopic Images for Melanoma Diagnosis," Intelligent Methods in Engineering Sciences, vol. 3, no. 2, pp. 70–81, 2024, doi: 10.58190/imiens.2024.101.*
- *S. Rama Krishna, A. Gudur, S. Jain, S. Deivasigamani, M. Tiwari, K. G. S. Venkatesan, and K. Raj V., "Deep Learning for Automatic Diagnosis of Skin Cancer Using Dermoscopic Images," Journal of Artificial Intelligence and Technology, vol. 4, no. 2, pp. 114–123, 2024, doi: 10.37965/jait.2024.0392.*

THANK YOU